UNIT 3: Mining Frequent Patterns, Associations, and Correlations: Basic and advanced Concepts

Basic Concepts, Frequent Itemset Mining Methods, Which Patterns Are Interesting? – Pattern Evaluation Methods, Pattern Mining: A Road Map, Pattern Mining in Multilevel, Multidimensional Space , Constraint-Based Frequent Pattern Mining, Pattern Exploration and Application.

Frequent pattern mining searches for recurring relationships in a given data set. This section introduces the basic concepts of frequent pattern mining for the discovery of interesting associations and correlations between itemsets in transactional and relational databases.

Frequent patterns are patterns (e.g., itemsets, subsequences, or substructures) that appear frequently in a data set. For example, a set of items, such as milk and bread, that appear frequently together in a transaction data set is a *frequent itemset*. A subsequence, such as buying first a PC, then a digital camera, and then a memory card, if it occurs frequently in a shopping history database, is a (*frequent*) *sequential pattern*. A *substructure* can refer to different structural forms, such as subgraphs, subtrees, or sublattices, which may be combined with itemsets or subsequences. If a substructure occurs frequently, it is called a (*frequent*) *structured pattern*. Finding frequent patterns plays an essential role in mining associations, correlations, and many other interesting relationships among data.

Moreover, it helps in data classification, clustering, and other data mining tasks. Thus, frequent pattern mining has become an important data mining task and a focused theme in data mining research.

Market Basket Analysis

A data mining technique that is used to uncover purchase patterns in any retail setting is known as **Market Basket Analysis**. In simple terms Basically, Market basket analysis in data mining is to analyze the combination of products which been bought together.

This is a technique that gives the careful study of purchases done by a customer in a supermarket. This concept identifies the pattern of frequent purchase items by customers. This analysis can help to promote deals, offers, sale by the companies, and data mining techniques helps to achieve this analysis task. Example:

- Data mining concepts are in use for Sales and marketing to provide better customer service, to improve cross-selling opportunities, to increase direct mail response rates.
- Customer Retention in the form of pattern identification and prediction of likely defections is possible by Data mining.
- Risk Assessment and Fraud area also use the data-mining concept for identifying inappropriate or unusual behavior etc.

Market basket analysis mainly works with the ASSOCIATION RULE {IF} -> {THEN}.

- **IF** means **Antecedent:** An antecedent is an item found within the data
- **THEN** means **Consequent:** A consequent is an item found in combination with the antecedent.



Let's see ASSOCIATION RULE {IF} -> {THEN} rules used in Market Basket Analysis in Data Mining. For example, customers buying a domain means they definitely need extra plugins/extensions to make it easier for the users.

Like we said above Antecedent is the item sets that are available in data. By formulating from the rules means **{if}** component and from the example is the domain.
Same as Consequent is the item that is found with the combination of Antecedents. By formulating from the rules means **{THEN}** component and from the example is extra plugins/extensions.
With the help of these, we are able to predict customer behavioral patterns. From this, we are able to make certain combinations with offers that customers will probably buy those products. That will automatically increase the sales and revenue of the company.

With the help of the Apriori Algorithm, we can further classify and simplify the item sets which are frequently bought by the consumer.
There are three components in APRIORI ALGORITHM:

- SUPPORT
- CONFIDENCE
- LIFT

Now take an example, suppose 5000 transactions have been made through a popular eCommerce website. Now they want to calculate the support, confidence, and lift for the two products, let's say pen and notebook for example out of 5000 transactions, 500 transactions for pen, 700 transactions for notebook, and 1000 transactions for both.

SUPPORT: It is been calculated with the number of transactions divided by the total number of transactions made, support(pen) = transactions related to pen/total transactions

i.e support -> 500/5000=10 percent

CONFIDENCE: It is been calculated for whether the product sales are popular on individual sales or through combined sales. That is calculated with combined transactions/individual transactions.
Confidence = combine transactions/individual transactions

i.e confidence-> 1000/500=20 percent

LIFT: Lift is calculated for knowing the ratio for the sales.
Lift-> 20/10=2

When the Lift value is below 1 means the combination is not so frequently bought by consumers. But in this case, it shows that the probability of buying both the things together is high when compared to the transaction for the individual items sold.

With this, we come to an overall view of the Market Basket Analysis in Data Mining and how to calculate the sales for combination products.

Types of Market Basket Analysis

There are three types of Market Basket Analysis. They are as follow:

1. **Descriptive market basket analysis**: This sort of analysis looks for patterns and connections in the data that exist between the components of a market basket. This kind of study is mostly used to understand consumer behavior, including what products are purchased in combination and what the most typical item combinations. Retailers can place products in their stores more profitably by understanding which products are frequently bought together with the aid of descriptive market basket analysis.
2. **Predictive Market Basket Analysis**: Market basket analysis that predicts future purchases based on past purchasing patterns is known as predictive market basket analysis. Large volumes of data are analyzed using machine learning algorithms in this sort of analysis in order to create predictions about which products are most likely to be bought together in the future. Retailers may make data-driven decisions about which products to carry, how to price them, and how to optimize shop layouts with the use of predictive market basket research.
3. **Differential Market Basket Analysis**: Differential market basket analysis analyses two sets of market basket data to identify variations between them. Comparing the behavior of various client segments or the behavior of customers over time is a common usage for this kind of study. Retailers can respond to shifting consumer behavior by modifying their marketing and sales tactics with the help of differential market basket analysis.

Benefits of Market Basket Analysis

1. **Enhanced Customer Understanding**: Market basket research offers insights into customer behavior, including what products they buy together and which products they buy the most frequently. Retailers can use this information to better understand their customers and make informed decisions.
2. **Improved Inventory Management**: By examining market basket data, retailers can determine which products are sluggish sellers and which ones are commonly bought together. Retailers can use this information to make well-informed choices about what products to stock and how to manage their inventory most effectively.
3. **Better Pricing Strategies**: A better understanding of the connection between product prices and consumer behavior might help merchants develop better pricing strategies. Using this knowledge, pricing plans that boost sales and profitability can be created.
4. **Sales Growth**: Market basket analysis can assist businesses in determining which products are most frequently bought together and where they should be positioned in the store to grow sales. Retailers may boost revenue and enhance customer shopping experiences by improving store layouts and product positioning.

Applications of Market Basket Analysis

1. **Retail**: Market basket research is frequently used in the retail sector to examine consumer buying patterns and inform decisions about product placement, inventory management, and pricing tactics. Retailers can utilize market basket research to identify which items are sluggish sellers and which ones are commonly bought together, and then modify their inventory management strategy accordingly.
2. **E-commerce**: Market basket analysis can help online merchants better understand the customer buying habits and make data-driven decisions about product recommendations and targeted advertising campaigns. The behaviour of visitors to a website can be examined using market basket analysis to pinpoint problem areas.
3. **Finance**: Market basket analysis can be used to evaluate investor behaviour and forecast the types of investment items that investors will likely buy in the future. The performance of investment portfolios can be enhanced by using this information to create tailored investment strategies.
4. **Telecommunications**: To evaluate consumer behaviour and make data-driven decisions about which goods and services to provide, the telecommunications business might employ market basket analysis. The usage of this data can enhance client happiness and the shopping experience.
5. **Manufacturing**: To evaluate consumer behaviour and make data-driven decisions about which products to produce and which materials to employ in the production process, the manufacturing sector might use market basket analysis. Utilizing this knowledge will increase effectiveness and cut costs.

A set of items is referred to as an **itemset.**

An itemset that contains *k* items is a ***k*-itemset**.

The set {*computer, antivirus software}* is a **2-itemset**.

The occurrence frequency of an itemset is the number of transactions that contain the itemset.

This is also known, simply, as the **frequency, support count,** or **count** of the itemset.

$$\begin{aligned} support(A \Rightarrow B) &= P(A \cup B) \\ confidence(A \Rightarrow B) &= P(B|A). \end{aligned}$$

$$confidence(A \Rightarrow B) = P(B|A) = \frac{support(A \cup B)}{support(A)} = \frac{support_count(A \cup B)}{support_count(A)}.$$

Rules that satisfy both a minimum support threshold (*min sup*) and a minimum confidence threshold (*min conf*) are called **Strong Association Rules.**

In general, association rule mining can be viewed as a **two-step process**:

1. Find all frequent itemsets: By definition, each of these itemsets will occur at least as frequently as a predetermined minimum support count, *min_sup*.
2. Generate strong association rules from the frequent itemsets: By definition, these rules must satisfy minimum support and minimum confidence.

Association Mining

Association rule mining: Finding frequent patterns, associations, correlations, or causal structures among sets of items or objects in transaction databases, relational databases, and other information repositories.

Applications: Basket data analysis, cross-marketing, catalog design, loss-leader analysis, clustering, classification, etc.

Frequent Itemset Mining Methods

1.Apriori algorithm

Apriori algorithm refers to the algorithm which is used to calculate the association rules between objects. It means how two or more objects are related to one another. In other words, we can say that the apriori algorithm is an association rule leaning that analyzes that people who bought product A also bought product B.

The primary objective of the apriori algorithm is to create the association rule between different objects. The association rule describes how two or more objects are related to one another. Apriori algorithm is also called frequent pattern mining. Generally, you operate the Apriori algorithm on a database that consists of a huge number of transactions. Let's understand the apriori algorithm with the help of an example; suppose you go to Big Bazar and buy different products. It helps the customers buy their products with ease and increases the sales performance of the Big Bazar. In this tutorial, we will discuss the apriori algorithm with examples.

Introduction

We take an example to understand the concept better. You must have noticed that the Pizza shop seller makes a pizza, soft drink, and breadstick combo together. He also offers a discount to their customers who buy these combos. Do you ever think why does he do so? He thinks that customers who buy pizza also buy soft drinks and breadsticks. However, by making combos, he makes it easy for the customers. At the same time, he also increases his sales performance.

Similarly, you go to Big Bazar, and you will find biscuits, chips, and Chocolate bundled together. It shows that the shopkeeper makes it comfortable for the customers to buy these products in the same place.

The above two examples are the best examples of Association Rules in Data Mining. It helps us to learn the concept of apriori algorithms.

What is Apriori Algorithm?

Apriori algorithm refers to an algorithm that is used in mining frequent products sets and relevant association rules. Generally, the apriori algorithm operates on a database containing a huge number of transactions. For example, the items customers but at a Big Bazar.

Apriori algorithm helps the customers to buy their products with ease and increases the sales performance of the particular store.

Components of Apriori algorithm

The given three components comprise the apriori algorithm.

1. Support
2. Confidence
3. Lift

Let's take an example to understand this concept.

We have already discussed above; you need a huge database containing a large no of transactions. Suppose you have 4000 customers transactions in a Big Bazar. You have to calculate the Support, Confidence, and Lift for two products, and you may say Biscuits and Chocolate. This is because customers frequently buy these two items together.

Out of 4000 transactions, 400 contain Biscuits, whereas 600 contain Chocolate, and these 600 transactions include a 200 that includes Biscuits and chocolates. Using this data, we will find out the support, confidence, and lift.

Support

Support refers to the default popularity of any product. You find the support as a quotient of the division of the number of transactions comprising that product by the total number of transactions. Hence, we get

Support (Biscuits) = (Transactions relating biscuits) / (Total transactions)

= 400/4000 = 10 percent.

Confidence

Confidence refers to the possibility that the customers bought both biscuits and chocolates together. So, you need to divide the number of transactions that comprise both biscuits and chocolates by the total number of transactions to get the confidence.

Hence,

Confidence = (Transactions relating both biscuits and Chocolate) / (Total transactions involving Biscuits)

= 200/400

= 50 percent.

It means that 50 percent of customers who bought biscuits bought chocolates also.

Lift

Consider the above example; lift refers to the increase in the ratio of the sale of chocolates when you sell biscuits. The mathematical equations of lift are given below.

Lift = (Confidence (Biscuits - chocolates)/ (Support (Biscuits)

= 50/10 = 5

It means that the probability of people buying both biscuits and chocolates together is five times more than that of purchasing the biscuits alone. If the lift value is below one, it requires that the people are unlikely to buy both the items together. Larger the value, the better is the combination.

How does the Apriori Algorithm work in Data Mining?

We will understand this algorithm with the help of an example

Consider a Big Bazar scenario where the product set is P = {Rice, Pulse, Oil, Milk, Apple}. The database comprises six transactions where 1 represents the presence of the product and 0 represents the absence of the product.

Transaction ID	Rice	Pulse	Oil	Milk	Apple
t1	1	1	1	0	0
t2	0	1	1	1	0
t3	0	0	0	1	1
t4	1	1	0	1	0

t5	1	1	1	0	1
t6	1	1	1	1	1

The Apriori Algorithm makes the given assumptions

- All subsets of a frequent itemset must be frequent.
- The subsets of an infrequent item set must be infrequent.
- Fix a threshold support level. In our case, we have fixed it at 50 percent.

Step 1

Make a frequency table of all the products that appear in all the transactions. Now, short the frequency table to add only those products with a threshold support level of over 50 percent. We find the given frequency table.

Product	Frequency (Number of transactions)
Rice (R)	4
Pulse(P)	5
Oil(O)	4
Milk(M)	4

The above table indicated the products frequently bought by the customers.

Step 2

Create pairs of products such as RP, RO, RM, PO, PM, OM. You will get the given frequency table.

Itemset	Frequency (Number of transactions)
RP	4

RO	3
RM	2
PO	4
PM	3
OM	2

Step 3

Implementing the same threshold support of 50 percent and consider the products that are more than 50 percent. In our case, it is more than 3

Thus, we get RP, RO, PO, and PM

Step 4

Now, look for a set of three products that the customers buy together. We get the given combination.

1. RP and RO give RPO
2. PO and PM give POM

Step 5

Calculate the frequency of the two itemsets, and you will get the given frequency table.

Itemset	Frequency (Number of transactions)
RPO	4
POM	3

If you implement the threshold assumption, you can figure out that the customers' set of three products is RPO.

We have considered an easy example to discuss the apriori algorithm in data mining. In reality, you find thousands of such combinations.

How to improve the efficiency of the Apriori Algorithm?

There are various methods used for the efficiency of the Apriori algorithm

Hash-based itemset counting

In hash-based itemset counting, you need to exclude the k-itemset whose equivalent hashing bucket count is least than the threshold is an infrequent itemset.

Transaction Reduction

In transaction reduction, a transaction not involving any frequent X itemset becomes not valuable in subsequent scans.

Apriori Algorithm in data mining

We have already discussed an example of the apriori algorithm related to the frequent itemset generation. Apriori algorithm has many applications in data mining.

The primary requirements to find the association rules in data mining are given below.

Use Brute Force

Analyze all the rules and find the support and confidence levels for the individual rule. Afterward, eliminate the values which are less than the threshold support and confidence levels.

The two-step approaches

The two-step approach is a better option to find the associations rules than the Brute Force method.

Step 1

In this article, we have already discussed how to create the frequency table and calculate itemsets having a greater support value than that of the threshold support.

Step 2

To create association rules, you need to use a binary partition of the frequent itemsets. You need to choose the ones having the highest confidence levels.

In the above example, you can see that the RPO combination was the frequent itemset. Now, we find out all the rules using RPO.

RP-O, RO-P, PO-R, O-RP, P-RO, R-PO

You can see that there are six different combinations. Therefore, if you have n elements, there will be 2^n - 2 candidate association rules.

Advantages of Apriori Algorithm

- It is used to calculate large itemsets.
- Simple to understand and apply.

Disadvantages of Apriori Algorithms

- Apriori algorithm is an expensive method to find support since the calculation has to pass through the whole database.
- Sometimes, you need a huge number of candidate rules, so it becomes computationally more expensive.
- In Data Mining, finding frequent patterns in large databases is very important and has been studied on a large scale in the past few years. Unfortunately, this task is computationally expensive, especially when many patterns exist.
- The FP-Growth Algorithm proposed by ***Han in***. This is an efficient and scalable method for mining the complete set of frequent patterns by pattern fragment growth, using an extended prefix-tree structure for storing compressed and crucial information about frequent patterns named frequent-pattern tree (FP-tree). In his study, Han proved that his method outperforms other popular methods for mining frequent patterns, e.g. the Apriori Algorithm and the TreeProjection. In some later works, it was proved that FP-Growth performs better than other methods, including ***Eclat*** and ***Relim***. The popularity and efficiency of the FP-Growth Algorithm contribute to many studies that propose variations to improve its performance.
- What is FP Growth Algorithm?
- The FP-Growth Algorithm is an alternative way to find frequent item sets without using candidate generations, thus improving performance. For so much, it uses a divide-and-conquer strategy. The core of this method is the usage of a special data structure named frequent-pattern tree (FP-tree), which retains the item set association information.
- **This algorithm works as follows:**
- First, it compresses the input database creating an FP-tree instance to represent frequent items.
- After this first step, it divides the compressed database into a set of conditional databases, each associated with one frequent pattern.
- Finally, each such database is mined separately.

Using this strategy, the FP-Growth reduces the search costs by recursively looking for short patterns and then concatenating them into the long frequent patterns.

In large databases, holding the FP tree in the main memory is impossible. A strategy to cope with this problem is to partition the database into a set of smaller databases (called projected databases) and then construct an FP-tree from each of these smaller databases.

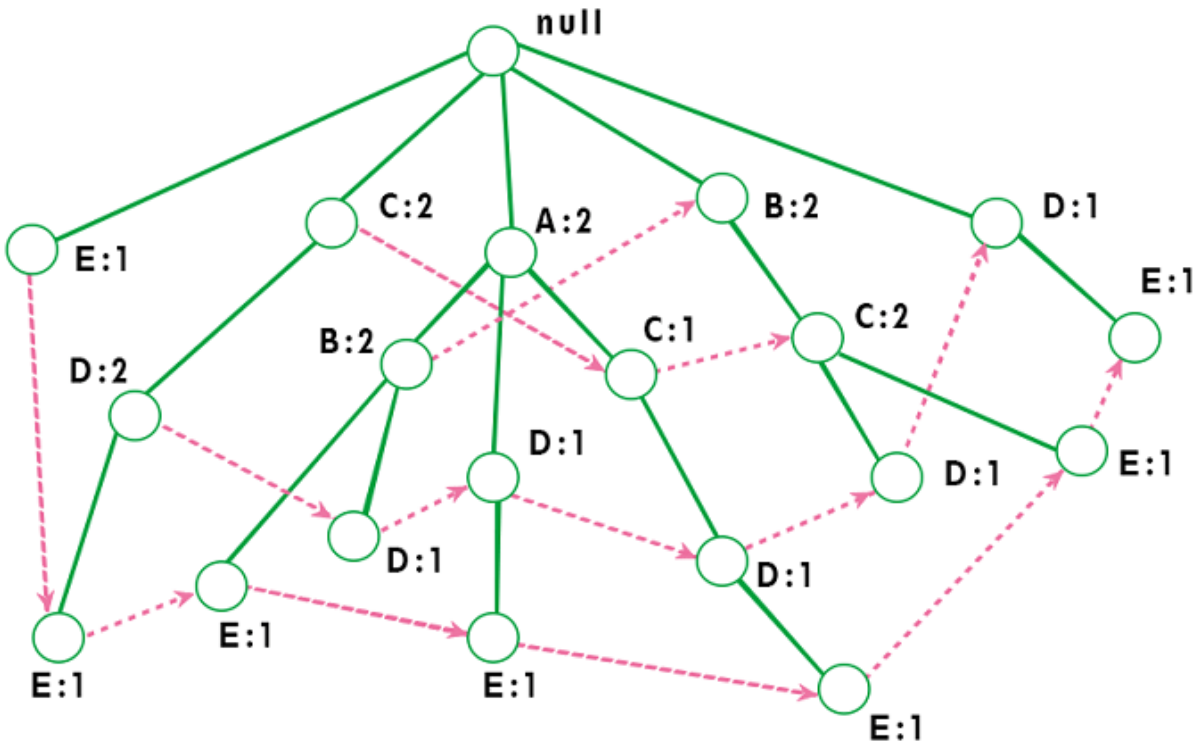
FP-Tree

The frequent-pattern tree (FP-tree) is a compact data structure that stores quantitative information about frequent patterns in a database. Each transaction is read and then mapped onto a path in the FP-tree. This is done until all transactions have been read. Different transactions with common subsets allow the tree to remain compact because their paths overlap.

A frequent Pattern Tree is made with the initial item sets of the database. The purpose of the FP tree is to mine the most frequent pattern. Each node of the FP tree represents an item of the item set.

The root node represents null, while the lower nodes represent the item sets. The associations of the nodes with the lower nodes, that is, the item sets with the other item sets, are maintained while forming the tree.

Han defines the FP-tree as the tree structure given below:



1. The first step is to scan the database to find the occurrences of the itemsets in the database. This step is the same as the first step of Apriori. The count of 1-itemsets in the database is called support count or frequency of 1-itemset.
2. The second step is to construct the FP tree. For this, create the root of the tree. The root is represented by null.
3. The next step is to scan the database again and examine the transactions. Examine the first transaction and find out the itemset in it. The itemset with the max count is taken at the top, and then the next itemset with the lower count. It means that the branch of the tree is constructed with transaction itemsets in descending order of count.
4. The next transaction in the database is examined. The itemsets are ordered in descending order of count. If any itemset of this transaction is already present in another branch, then this transaction branch would share a common prefix to the root.
   This means that the common itemset is linked to the new node of another itemset in this transaction.
5. Also, the count of the itemset is incremented as it occurs in the transactions. The common node and new node count are increased by 1 as they are created and linked according to transactions.
6. The next step is to mine the created FP Tree. For this, the lowest node is examined first, along with the links of the lowest nodes. The lowest node represents the frequency pattern length 1. From this, traverse the path in the FP Tree. This path or paths is called a conditional pattern base.
   A conditional pattern base is a sub-database consisting of prefix paths in the FP tree occurring with the lowest node (suffix).
7. Construct a Conditional FP Tree, formed by a count of itemsets in the path. The itemsets meeting the threshold support are considered in the Conditional FP Tree.
8. Frequent Patterns are generated from the Conditional FP Tree.

Using this algorithm, the FP-tree is constructed in two database scans. The first scan collects and sorts the set of frequent items, and the second constructs the FP-Tree.

Example

Support threshold=50%, Confidence= 60%

Table 1:

Transaction	List of items
T1	I1,I2,I3
T2	I2,I3,I4
T3	I4,I5
T4	I1,I2,I4
T5	I1,I2,I3,I5
T6	I1,I2,I3,I4

Solution: Support threshold=50% => 0.5*6= 3 => min_sup=3

Table 2: Count of each item

Item	Count
I1	4
I2	5
I3	4
I4	4
I5	2

Table 3: Sort the itemset in descending order.

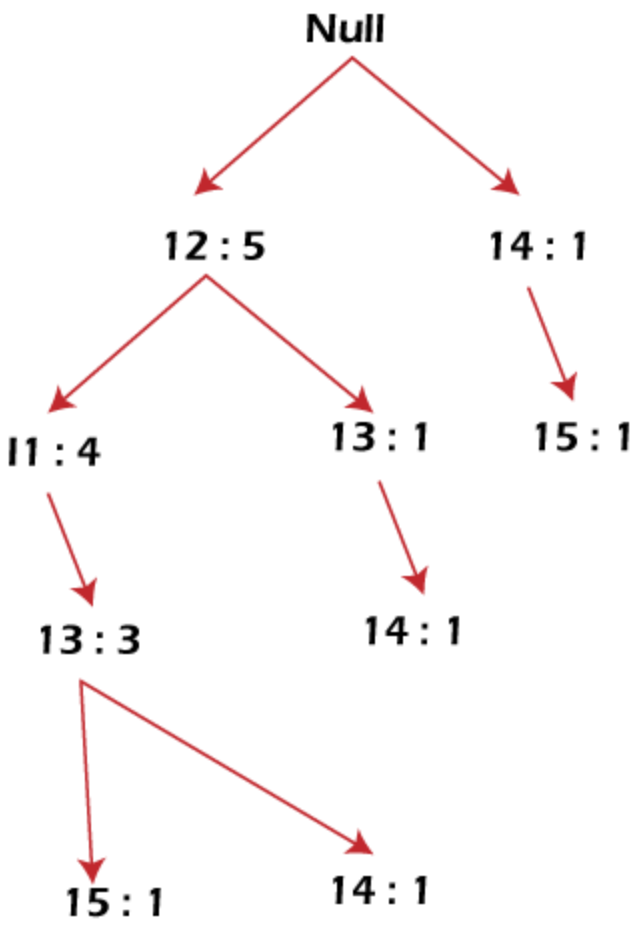
Item	Count

I2	5
I1	4
I3	4
I4	4

Build FP Tree

Let's build the FP tree in the following steps, such as:

1. Considering the root node null.
2. The first scan of Transaction T1: I1, I2, I3 contains three items {I1:1}, {I2:1}, {I3:1}, where I2 is linked as a child, I1 is linked to I2 and I3 is linked to I1.
3. T2: I2, I3, and I4 contain I2, I3, and I4, where I2 is linked to root, I3 is linked to I2 and I4 is linked to I3. But this branch would share the I2 node as common as it is already used in T1.
4. Increment the count of I2 by 1, and I3 is linked as a child to I2, and I4 is linked as a child to I3. The count is {I2:2}, {I3:1}, {I4:1}.
5. T3: I4, I5. Similarly, a new branch with I5 is linked to I4 as a child is created.
6. T4: I1, I2, I4. The sequence will be I2, I1, and I4. I2 is already linked to the root node. Hence it will be incremented by 1. Similarly I1 will be incremented by 1 as it is already linked with I2 in T1, thus {I2:3}, {I1:2}, {I4:1}.
7. T5:I1, I2, I3, I5. The sequence will be I2, I1, I3, and I5. Thus {I2:4}, {I1:3}, {I3:2}, {I5:1}.
8. T6: I1, I2, I3, I4. The sequence will be I2, I1, I3, and I4. Thus {I2:5}, {I1:4}, {I3:3}, {I4 1}.



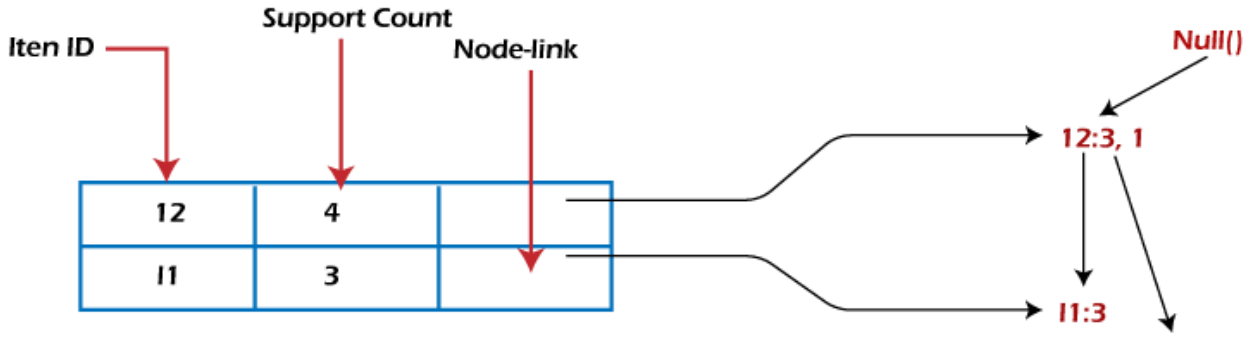
Mining of FP-tree is summarized below:

1. The lowest node item, I5, is not considered as it does not have a min support count. Hence it is deleted.
2. The next lower node is I4. I4 occurs in 2 branches , {I2,I1,I3:,I41},{I2,I3,I4:1}. Therefore considering I4 as suffix the prefix paths will be {I2, I1, I3:1}, {I2, I3: 1} this forms the conditional pattern base.
3. The conditional pattern base is considered a transaction database, and an FP tree is constructed. This will contain {I2:2, I3:2}, I1 is not considered as it does not meet the min support count.
4. This path will generate all combinations of frequent patterns : {I2,I4:2},{I3,I4:2},{I2,I3,I4:2}
5. For I3, the prefix path would be: {I2,I1:3},{I2:1}, this will generate a 2 node FP-tree : {I2:4, I1:3} and frequent patterns are generated: {I2,I3:4}, {I1:I3:3}, {I2,I1,I3:3}.

6. For I1, the prefix path would be: {I2:4} this will generate a single node FP-tree: {I2:4} and frequent patterns are generated: {I2, I1:4}.

Item	Conditional Pattern Base	Conditional FP-tree	Frequent Patterns Generated
I4	{I2,I1,I3:1},{I2,I3:1}	{I2:2, I3:2}	{I2,I4:2},{I3,I4:2},{I2,I3,I4:2}
I3	{I2,I1:3},{I2:1}	{I2:4, I1:3}	{I2,I3:4}, {I1:I3:3}, {I2,I1,I3:3}
I1	{I2:4}	{I2:4}	{I2,I1:4}

The diagram given below depicts the conditional FP tree associated with the conditional node I3.



FP-Growth Algorithm

After constructing the FP-Tree, it's possible to mine it to find the complete set of frequent patterns. Han presents a group of lemmas and properties to do this job and then describes the following FP-Growth Algorithm.

Advantages of FP Growth Algorithm

Here are the following advantages of the FP growth algorithm, such as:

- This algorithm needs to scan the database twice when compared to Apriori, which scans the transactions for each iteration.
- The pairing of items is not done in this algorithm, making it faster.
- The database is stored in a compact version in memory.

- It is efficient and scalable for mining both long and short frequent patterns.

Disadvantages of FP-Growth Algorithm

This algorithm also has some disadvantages, such as:

- FP Tree is more cumbersome and difficult to build than Apriori.
- It may be expensive.
- The algorithm may not fit in the shared memory when the database is large.

Difference between Apriori and FP Growth Algorithm

Apriori and FP-Growth algorithms are the most basic FIM algorithms. There are some basic differences between these algorithms, such as:

Apriori	**FP Growth**
Apriori generates frequent patterns by making the itemsets using pairings such as single item set, double itemset, and triple itemset.	FP Growth generates an FP-Tree for making frequent patterns.
Apriori uses candidate generation where frequent subsets are extended one item at a time.	FP-growth generates a conditional FP-Tree for every item in the data.
Since apriori scans the database in each step, it becomes time-consuming for data where the number of items is larger.	FP-tree requires only one database scan in its beginning steps, so it consumes less time.
A converted version of the database is saved in the memory	A set of conditional FP-tree for every item is saved in the memory
It uses a breadth-first search	It uses a depth-first search.

Which Patterns Are Interesting?—Pattern Evaluation Methods

The support and confidence measures are insufficient at filtering out uninteresting association rules. To tackle this weakness, a correlation measure can be used to augment the support–confidence framework for association rules. This leads to *correlation rules* of the form

$A \rightarrow B$ [*support*, *confidence*, *correlation*].

That is, a correlation rule is measured not only by its support and confidence but also by the correlation between itemsets *A* and *B*. There are many different correlation measures from which to choose.
In data mining, pattern evaluation is the process of assessing the quality of discovered patterns. This process is important in order to determine whether the patterns are useful and whether they can be trusted. There are a number of different measures that can be used to evaluate patterns, and the choice of measure will depend on the application.

There are several **ways to evaluate pattern mining algorithms:**

1. Accuracy

The accuracy of a data mining model is a measure of how correctly the model predicts the target values. The accuracy is measured on a test dataset, which is separate from the training dataset that was used to train the model. There are a number of ways to measure accuracy, but the most common is to calculate the percentage of correct predictions. This is known as the accuracy rate.

Other measures of accuracy include the root mean squared error (RMSE) and the mean absolute error (MAE). The RMSE is the square root of the mean squared error, and the MAE is the mean of the absolute errors. The accuracy of a data mining model is important, but it is not the only thing that should be considered. The model should also be robust and generalizable.

A model that is 100% accurate on the training data but only 50% accurate on the test data is not a good model. The model is overfitting the training data and is not generalizable to new data. A model that is 80% accurate on the training data and 80% accurate on the test data is a good model. The model is generalizable and can be used to make predictions on new data.

2. Classification Accuracy

This measures how accurately the patterns discovered by the algorithm can be used to classify new data. This is typically done by taking a set of data that has been labeled with known class labels and then using the discovered patterns to predict the class labels of the data. The accuracy can then be computed by comparing the predicted labels to the actual labels.

Classification accuracy is one of the most popular evaluation metrics for classification models, and it is simply the percentage of correct predictions made by the model. Although it is a straightforward and easy-to-understand metric, classification accuracy can be misleading in certain situations. **For example,** if we have a dataset with a very imbalanced class distribution, such as 100 instances of class 0 and 1,000 instances of class 1, then a model that always predicts class 1 will achieve a high classification accuracy of 90%. However, this model is clearly not very useful, since it is not making any correct predictions for class 0.
There are a few different ways to evaluate classification models, such as precision and recall, which are more informative in imbalanced datasets. **Precision** is the percentage of correct

predictions made by the model for a particular class, and **recall** is the percentage of instances of a particular class that was correctly predicted by the model. In the above **example,** if we looked at precision and recall for class 0, we would see that the model has a precision of 0% and a recall of 0%.
Another way to evaluate classification models is to use a confusion matrix. A confusion matrix is a table that shows the number of correct and incorrect predictions made by the model for each class. This can be a helpful way to visualize the performance of a model and to identify where it is making mistakes. For example, in the above example, the confusion matrix would show that the model is making all predictions for class 1 and no predictions for class 0.

Overall, classification accuracy is a good metric to use when evaluating classification models. However, it is important to be aware of its limitations and to use other evaluation metrics in situations where classification accuracy could be misleading.

3. Clustering Accuracy
This measures how accurately the patterns discovered by the algorithm can be used to cluster new data. This is typically done by taking a set of data that has been labeled with known cluster labels and then using the discovered patterns to predict the cluster labels of the data. The accuracy can then be computed by comparing the predicted labels to the actual labels.

There are a few **ways to evaluate the accuracy of a clustering algorithm:**

- **External indices:** these indices compare the clusters produced by the algorithm to some known ground truth. For example, the Rand Index or the Jaccard coefficient can be used if the ground truth is known.
- **Internal indices:** these indices assess the goodness of clustering without reference to any external information. The most popular internal index is the Dunn index.
- **Stability:** this measures how robust the clustering is to small changes in the data. A clustering algorithm is said to be stable if, when applied to different samples of the same data, it produces the same results.
- **Efficiency:** this measures how quickly the algorithm converges to the correct clustering.

4. Coverage
This measures how many of the possible patterns in the data are discovered by the algorithm. This can be computed by taking the total number of possible patterns and dividing it by the number of patterns discovered by the algorithm. A Coverage Pattern is a type of sequential pattern that is found by looking for items that tend to appear together in sequential order. For **example,** a coverage pattern might be "customers who purchase item A also tend to purchase item B within the next month."
To evaluate a coverage pattern, analysts typically look at two things: support and confidence. Support is the percentage of transactions that contain the pattern. Confidence is the percentage of transactions that contain the pattern divided by the number of transactions that contain the first item in the pattern.

For **example,** consider the following coverage pattern: "customers who purchase item A also tend to purchase item B within the next month." If the support for this pattern is 0.1%, that means that 0.1% of all transactions contain the pattern. If the confidence for this pattern is 80%, that means that 80% of the transactions that contain item A also contain item B.

Generally, a higher support and confidence value indicates a stronger pattern. However, analysts must be careful to avoid overfitting, which is when a pattern is found that is too specific to the data and would not be generalizable to other data sets.

5. Visual Inspection

This is perhaps the most common method, where the data miner simply looks at the patterns to see if they make sense. In visual inspection, the data is plotted in a graphical format and the pattern is observed. This method is used when the data is not too large and can be easily plotted. It is also used when the data is categorical in nature. Visual inspection is a pattern evaluation method in data mining where the data is visually inspected for patterns. This can be done by looking at a graph or plot of the data, or by looking at the raw data itself. This method is often used to find outliers or unusual patterns.

6. Running Time

This measures how long it takes for the algorithm to find the patterns in the data. This is typically measured in seconds or minutes. There are a few different ways to measure the performance of a machine learning algorithm, but one of the most common is to simply measure the amount of time it takes to train the model and make predictions. This is known as the running time pattern evaluation.

There are a few different things to keep in mind when measuring the running time of an algorithm. First, you need to take into account the time it takes to load the data into memory. Second, you need to account for the time it takes to pre-process the data if any. Finally, you need to account for the time it takes to train the model and make predictions.

In general, the running time of an algorithm will increase as the number of data increases. This is because the algorithm has to process more data in order to learn from it. However, there are some algorithms that are more efficient than others and can scale to large datasets better. When comparing different algorithms, it is important to keep in mind the specific dataset that is being used. Some algorithms may be better suited for certain types of data than others. In addition, the running time can also be affected by the hardware that is being used.

7. Support

The support of a pattern is the percentage of the total number of records that contain the pattern. Support Pattern evaluation is a process of finding interesting and potentially useful patterns in data. The purpose of support pattern evaluation is to identify interesting patterns that may be useful for decision-making. Support pattern evaluation is typically used in data mining and machine learning applications.

There are a variety of ways to evaluate support patterns. One common approach is to use a support metric, which measures the number of times a pattern occurs in a dataset. Another common approach is to use a lift metric, which measures the ratio of the occurrence of a pattern to the expected occurrence of the pattern.

Support pattern evaluation can be used to find a variety of interesting patterns in data, including association rules, sequential patterns, and co-occurrence patterns. Support pattern evaluation is an important part of data mining and machine learning, and can be used to help make better decisions.

8. Confidence

The confidence of a pattern is the percentage of times that the pattern is found to be correct. Confidence Pattern evaluation is a method of data mining that is used to assess the quality of patterns found in data. This evaluation is typically performed by calculating the percentage of times a pattern is found in a data set and comparing this percentage to the percentage of times the pattern is expected to be found based on the overall distribution of data. If the percentage of times a pattern is found is significantly higher than the expected percentage, then the pattern is said to be a strong confidence pattern.

9. Lift

The lift of a pattern is the ratio of the number of times that the pattern is found to be correct to the number of times that the pattern is expected to be correct. Lift Pattern evaluation is a data mining technique that can be used to evaluate the performance of a predictive model. The lift pattern is a graphical representation of the model's performance and can be used to identify potential problems with the model.

The lift pattern is a plot of the true positive rate (TPR) against the false positive rate (FPR). The TPR is the percentage of positive instances that are correctly classified by the model, while the FPR is the percentage of negative instances that are incorrectly classified as positive. Ideally, the TPR would be 100% and the FPR would be 0%, but this is rarely the case in practice. The lift pattern can be used to evaluate how close the model is to this ideal.

A good model will have a lifted pattern that is close to the diagonal line. This means that the TPR and FPR are similar and that the model is correctly classifying a similar percentage of positive and negative instances. A model with a lifted pattern that is far from the diagonal line is not performing as well. This can be caused by a number of factors, including imbalanced data, poor feature selection, or overfitting.

The lift pattern can be a useful tool for identifying potential problems with a predictive model. It is important to remember, however, that the lift pattern is only a graphical representation of the model's performance, and should be interpreted in conjunction with other evaluation measures.

10. Prediction

The prediction of a pattern is the percentage of times that the pattern is found to be correct. Prediction Pattern evaluation is a data mining technique used to assess the accuracy of predictive models. It is used to determine how well a model can predict future outcomes based on past data. Prediction Pattern evaluation can be used to compare different models, or to evaluate the performance of a single model.

Prediction Pattern evaluation involves splitting the data set into two parts: a training set and a test set. The training set is used to train the model, while the test set is used to assess the accuracy of the model. To evaluate the accuracy of the model, the prediction error is calculated. Prediction Pattern evaluation can be used to improve the accuracy of predictive models. By using a test set, predictive models can be fine-tuned to better fit the data. This can be done by changing the model parameters or by adding new features to the data set.

11. Precision

Precision Pattern Evaluation is a method for analyzing data that has been collected from a variety of sources. This method can be used to identify patterns and trends in the data, and to evaluate the accuracy of data. Precision Pattern Evaluation can be used to identify errors in the data, and to determine the cause of the errors. This method can also be used to determine the impact of the errors on the overall accuracy of the data.

Precision Pattern Evaluation is a valuable tool for data mining and data analysis. This method can be used to improve the accuracy of data, and to identify patterns and trends in the data.

12. Cross-Validation

This method involves partitioning the data into two sets, training the model on one set, and then testing it on the other. This can be done multiple times, with different partitions, to get a more reliable estimate of the model's performance. Cross-validation is a model validation technique for assessing how the results of a data mining analysis will generalize to an independent data set. It is mainly used in settings where the goal is prediction, and one wants to estimate how accurately a predictive model will perform in practice. Cross-validation is also referred to as out-of-sample testing.

Cross-validation is a pattern evaluation method that is used to assess the accuracy of a model. It does this by splitting the data into a training set and a test set. The model is then fit on the training set and the accuracy is measured on the test set. This process is then repeated a number of times, with the accuracy being averaged over all the iterations.

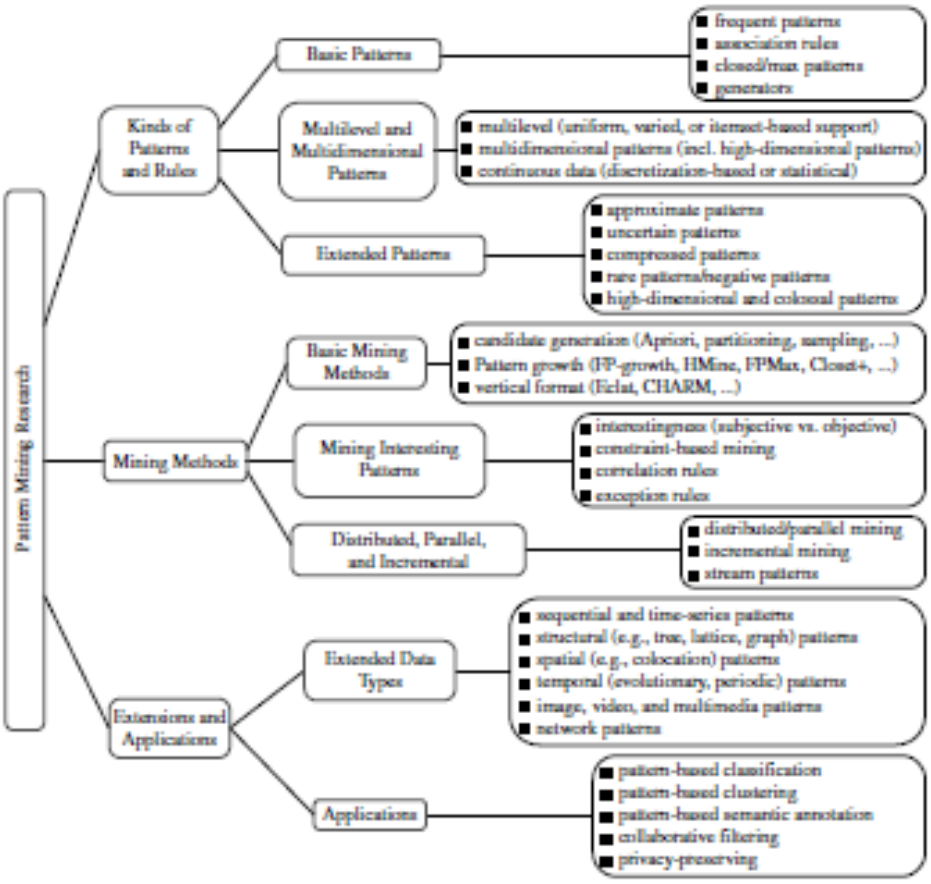
13. Test Set

This method involves partitioning the data into two sets, training the model on the entire data set, and then testing it on the held-out test set. This is more reliable than cross-validation but can be more expensive if the data set is large. There are a number of ways to evaluate the performance of a model on a test set. The most common is to simply compare the predicted labels to the true labels and compute the percentage of instances that are correctly classified. This is called accuracy. Another popular metric is precision, which is the number of true positives divided by the sum of true positives and false positives. The recall is the number of true positives divided by the sum of true positives and false negatives. These metrics can be combined into the F1 score, which is the harmonic mean of precision and recall.

14. Bootstrapping

This method involves randomly sampling the data with replacement, training the model on the sampled data, and then testing it on the original data. This can be used to get a distribution of the model's performance, which can be useful for understanding how robust the model is. Bootstrapping is a resampling technique used to estimate the accuracy of a model. It involves randomly selecting a sample of data from the original dataset and then training the model on this sample. The model is then tested on another sample of data that is not used in training. This process is repeated a number of times, and the average accuracy of the model is calculated.

Pattern Mining: A Road Map



Based on pattern diversity, pattern mining can be classified using the following criteria:

Basic patterns: As discussed in Chapter 6, a frequent pattern may have several alternative forms, including a simple frequent pattern, a closed pattern, or a max-pattern. To review, a **frequent pattern** is a pattern (or itemset) that satisfies a minimum support threshold. A pattern *p* is a **closed pattern** if there is no super pattern *p*0 with the same support as *p*. Pattern *p* is a **max-pattern** if there exists no frequent super pattern of *p*. Frequent patterns can also be mapped into **association rules**, or other kinds of rules based on interestingness measures. Sometimes we may also be interested in **infrequent** or **rare patterns** (i.e., patterns that occur rarely but are of critical importance, or **negative patterns** (i.e., patterns that reveal a negative correlation between items).

Based on the *abstraction* levels involved in a pattern: Patterns or association rules may have items or concepts residing at high, low, or multiple abstraction levels. For example, suppose that a set of association rules mined includes the following rules where *X* is a variable representing a customer: *buys.X*, "*computer*"/)*buys.X*, "*printer* "/ (1)
buys.X, "*laptop computer*"/)*buys.X*, "*color laser printer* "/ (2)
In Rules (1) and (2), the items bought are referenced at different abstraction levels (e.g., "*computer*" is a higher-level abstraction of "*laptop computer*," and "*color laser printer*" is a lower-

level abstraction of "*printer*"). We refer to the rule set mined as consisting of **multilevel association rules**. If, instead, the rules within a given set do not reference items or attributes at different abstraction levels, then the set contains **single-level association rules**.

Based on the *number of dimensions* involved in the rule or pattern: If the items or attributes in an association rule or pattern reference only one dimension, it is a **single-dimensional association rule/pattern**. For example, Rules (1) and (2) are single-dimensional association rules because they each refer to only one dimension, *buys*.1

If a rule/pattern references two or more dimensions, such as *age*, *income*, and *buys*, then it is a **multidimensional association rule/pattern**. The following is an example of a multidimensional rule:

age.X, "20: : :29"/^*income.X*, "52*K* : : :58*K*"/)*buys.X*, "*iPad* "/. (3)

Based on the *types of values* handled in the rule or pattern: If a rule involves associations between the presence or absence of items, it is a **Boolean association rule**. For example, Rules (1) and (2) are Boolean association rules obtained from market basket analysis.

If a rule describes associations between quantitative items or attributes, then it is a **quantitative association rule**. In these rules, quantitative values for items or attributes are partitioned into intervals. Rule (3) can also be considered a quantitative association rule where the quantitative attributes *age* and *income* have been discretized.

Based on the *constraints* or *criteria* used to mine *selective patterns*: The patterns or rules to be discovered can be **constraint-based** (i.e., satisfying a set of user defined constraints), **approximate**, **compressed**, **near-match** (i.e., those that tally the support count of the near or almost matching itemsets), **top-*k*** (i.e., the *k* most frequent itemsets for a user-specified value, *k*), **redundancy-aware top-*k*** (i.e., the top-*k* patterns with similar or redundant patterns excluded), and so on.

Alternatively, pattern mining can be classified with respect to the kinds of data and applications involved, using the following criteria:

Based on *kinds of data and features* to be mined: Given relational and data warehouse data, most people are interested in itemsets. Thus, frequent pattern mining in this context is essentially **frequent itemset mining**, that is, to mine frequent *sets of items*. However, in many other applications, patterns may involve sequences and structures. For example, by studying the order in which items are frequently purchased, we may find that customers tend to first buy a PC, followed by a digital camera, and then a memory card. This leads to **sequential patterns**, that is, frequent *subsequences* (which are often separated by some other events) in a *sequence of ordered events*.

We may also mine **structural patterns**, that is, frequent *substructures*, in a *structured data set*. Note that *structure* is a general concept that covers many different kinds of structural forms such as directed graphs, undirected graphs, lattices, trees, sequences, sets, single items, or combinations of such structures. Single items are the simplest form of structure. Each element of a general pattern may contain a subsequence, a subtree, a subgraph, and so on, and such containment relationships can be defined recursively. Therefore, structural pattern mining can be considered as the most general form of frequent pattern mining.

Based on *application domain-specific semantics*: Both data and applications can be very diverse, and therefore the patterns to be mined can differ largely based on their domain-specific semantics. Various kinds of application data include spatial data, temporal data, spatiotemporal data, multimedia data (e.g., image, audio, and video data), text data, time-series data, DNA and biological sequences, software programs, chemical compound structures, web structures, sensor networks, social and information networks, biological networks, data streams, and so on. This diversity can lead to dramatically different pattern mining methodologies.

Based on *data analysis usages*: Frequent pattern mining often serves as an intermediate step for improved data understanding and more powerful data analysis. For example, it can be used as a feature extraction step for classification, which is often referred to as **pattern-based classification**. Similarly, **pattern-based clustering** has shown its strength at clustering high-dimensional data. For improved data understanding, patterns can be used for semantic annotation or contextual analysis. Pattern analysis can also be used in **recommender systems**, which recommend information items (e.g., books, movies, web pages) that are likely to be of interest to the user based on similar users' patterns. Different analysis tasks may require mining rather different kinds of patterns as well.

Pattern Mining in Multilevel, Multidimensional Space

Association Rule in Data Mining

Strong correlations can be found at high levels of abstraction, even with great support; nonetheless, for many applications, these correlations can be regarded as common sense. There is a possibility that we will need to delve much farther into the data to recognize patterns that are genuinely one of a kind.

On the other hand, there may be many patterns at elementary abstraction levels, many of which are trivializations of patterns at higher degrees of abstraction. Consequently, it is fascinating to contemplate how to construct effective algorithms for mining patterns across a wide range of abstraction levels while ensuring that these algorithms have adequate flexibility to permit easy travel between these areas.

Example of Association Rule in Data Mining

You can make data more generalizable by replacing certain instances of low-level concepts with their higher-level concept relatives, also known as ancestors, from a concept hierarchy. This will make the data more generic.

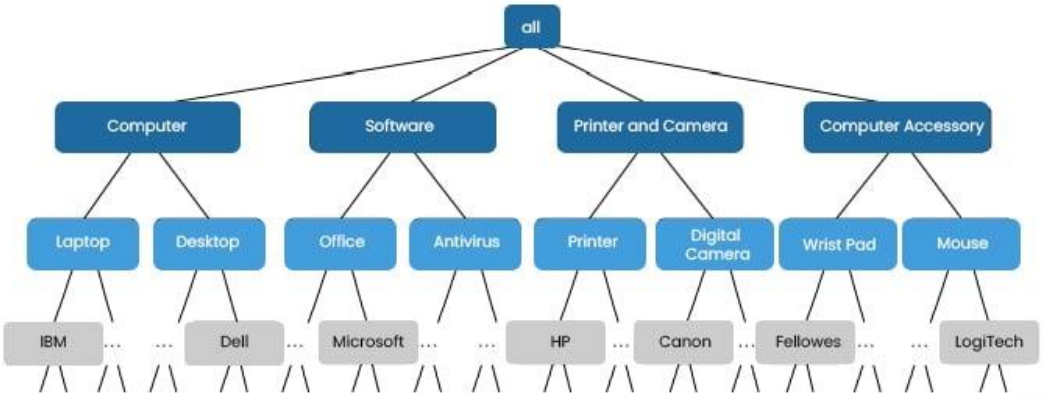
Example:

Let's imagine we have access to the critically crucial collection of sales data from an electronics store, displayed in Table 1. The ideas located at the bottom of an order are the ones that have the least generalization, while the concepts situated at the very top are the ones that are the most particular.

Task-Relevant Data, D

TID	Items Purchased
T100	Apple 17" MacBook Pro Notebook, HP Photosmart Pro b9180
T200	Microsoft Office Professional 2010, Microsoft Wireless Optical Mouse 5000
T300	Logitech VX Nano Cordless Laser Mouse, Fellowes GEL Wrist Rest
T400	Dell Studio XPS 16 Notebook, Canon PowerShot SD1400
T500	Lenovo ThinkPad X200 Tablet PC, Symantec Norton Antivirus 2010

In the order of five levels of concept, representation depicted, "Degree 0" stands for the most fundamental degree of abstraction. The first three levels are considered intermediate degrees of abstraction, while level 4 is considered the most significant level of abstraction. In this scenario, level 1 includes items such as computers, software, printers, cameras, and other computer accessories; level 2 includes laptops and desktops, as well as office and antivirus software; and level 3, includes high-end brands such as Dell desktops, Microsoft office software, and other similar products. Considering this hierarchy, the level 4 abstraction represents the most feasible detailed level. The numbers that are contained in the raw data have been incorporated.



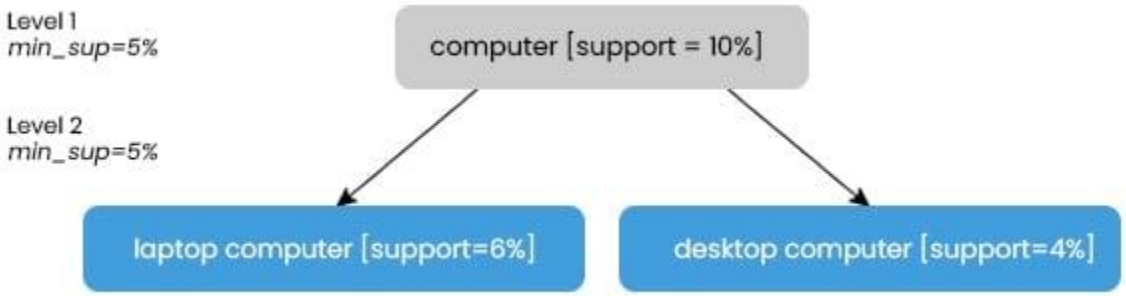
Nominal attribute concept hierarchies are typically already present in the database schema. They can be automatically constructed with techniques such as those outlined. Data on product requirements were used to create the concept hierarchy for our example. Many discretization techniques were discussed to develop concept hierarchies for numeric characteristics. Users familiar with the data, like store managers in our example, may also provide idea hierarchies.

Finding engaging purchasing trends with such raw or basic data is challenging. Finding significant correlations with specific goods might be difficult if they occur in a tiny percentage of transactions, such as the "*Dell Studio XPS 16 Notebook*" or the "*Logitech VX Nano Cordless Laser Mouse*." Due to low expected group buying, this bundle is unlikely to reach critical mass.

In contrast, we anticipate that it is simpler to discover robust correlations between broad categories of these things, such as "Dell Notebook" and "Cordless Mouse."

Multilevel Association Rule Mining

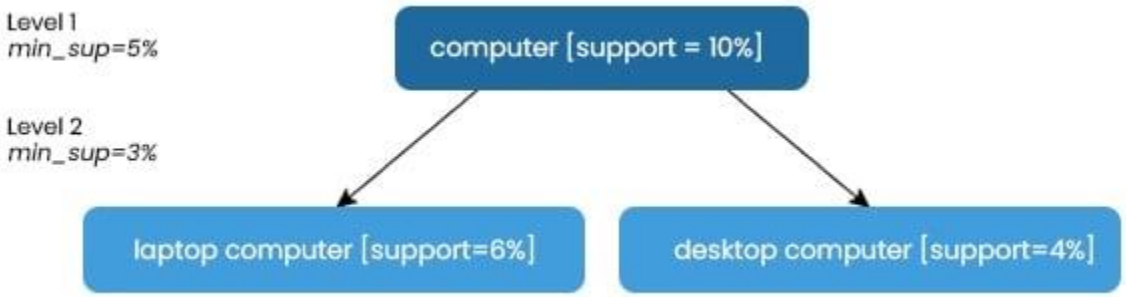
Many-level, or multilevel, association rules are a type of association rule developed via data mining at multiple abstraction levels. Rules of the association at several levels are mined effectively by employing idea hierarchies inside a support-confidence framework. Frequent item sets are often calculated using a top-down approach, where counts are amassed for each idea level from the most general to the most particular, and so on, until no more frequent item sets can be identified. For each tier, you can use Apriori or any other method for finding frequently occurring item sets.



Methods for Association Rule Mining:

What follows is a discussion of many alternatives to this method, each of which requires "playing" with the support threshold somewhat differently. Nodes represent things or item sets that have been investigated, and nodes with thick margins represent items or set examinations that have occurred often.

We have uniform support by providing the same amount of backing across the board.



- Minimal Support Criterion Method

Mining at each level of abstraction uses the same minimal support criterion.

A minimum support requirement of 5% is applied (for mining from "*computer*" down to "*laptop computer*"). The terms "*computer*" and "*laptop computer*" are standard, whereas "*desktop computer*" is rarely used.

Using a standard minimum support level can streamline the search process. The process is extremely straightforward because consumers need minimal support from only one. Given that every ancestor is a superset of its descendants, we may use an Apriori-like optimization strategy:

The search will ignore any item for which the ancestors do not have minimal support.

Despite its popularity, the uniform support method has its challenges. Products with a greater degree of abstraction are more likely to appear frequently than those with a lower level of abstraction. Because of this, it's essential to find a happy medium between setting the minimum support threshold high enough to catch all but the most vital links and making it too low to miss any significant ones. Too low of a point might cause the generation of numerous irrelevant linkages at very high abstraction levels.

- Reduced Support Method

This is the impetus for the following strategy. Through the use of lower levels of minimal support (hence referred to as "*reduced support*"):

A lower threshold of support is required at lower levels of abstraction. The threshold is lower at higher levels of abstraction. The above figure shows that 5% and 3% minimum support are required for levels 1 and 2, respectively. Computer, laptop computer, and desktop computer are all examples of common noun phrases here.

Building up user-specific, item- or group-based minimal support levels may be preferable when mining multilayer rules, as users or experts typically have insight into which groups are more significant than others. Set minimum support levels depending on the product price or items of interest; for instance, pay special attention to the association patterns, including those categories (such as "camera with price over $1000" or "Tablet PC") and the system will prioritize such patterns.

The threshold is often set to the lowest value among the participating groups for mining patterns containing elements from many groups with varying support thresholds. This way, useful patterns comprising elements from the least-supported group won't be filtered out. Meanwhile, it's important to maintain the minimum required level of support for each group to prevent the accumulation of boring collections of things from any group. After the itemset mining, different metrics of interestingness may be applied to draw out the most pertinent rules.

When mining under limited or group-based support, remember that the Apriori attribute does not always hold evenly across all objects. On the other hand, effective strategies may be created by expanding the property.

As a negative byproduct of the "*ancestor*" relationships among things, pattern mining multilevel association rules generates numerous duplicate rules across several abstraction levels. Using the idea hierarchy, one can see that the "laptop computer" is the progenitor of "Dell laptop computer," prompting us to evaluate the following rules:

Clients who made purchases in AllElectronics transactions are represented by the variable X

Support = 8%, Confidence = 70% buys(X, "laptop computer") > buys(X, "HP printer").

compare('buys(X, 'Dell laptop computer') > compare('buys(X, 'HP printer')] [support = 2%, confi dence = 72%] give something new or different?" We should eliminate the latter, less broad rule if it doesn't add anything to the conversation. Let's consider some possible methods of determining this. If the items in R2 can be acquired by replacing them with their predecessors in a concept hie rarchy, then R2 is an ancestor of R1.

The reason is that "*laptop computer*" predates "Dell laptop computer." This definition suggests that a rule is redundant if its support and confidence are near the "*anticipated*" values based on a rule's parent.

Multidimensional Associations Rule in Data Mining

Association rules with a single implying predicate, such as "*buys*," have been the focus of our attention so far. The Boolean association rule buys(X, "digital camera") buys(X, "HP printer"), for instance, may be found by mining our AllElectronics database.

Each unique predicate in a rule is referred to as a dimension, in keeping with the terminology used in multidimensional databases. Thus, the Rule is a single-dimensional or intra-dimensional association rule since it has a single distinct predicate (such as buys) with multiple occurrences (i.e., the predicate appears more than once within the rule). Rules like this may often be extracted from transactional data. You can learn more about six stages of data science processing to grasp the above topic better.

When analyzing sales and related information, it's not uncommon to combine it with relational data or include it in a data warehouse rather than relying just on transactional data. Multidimensionality is a hallmark of such data warehouses. For instance, a relational database may store additional information linked with sales transactions outside just the things purchased.

Associated with the goods and the deal itself, including the description of the goods or the location of the sale's branch. A customer's age, profession, credit score, income, and address are all examples of relational data that may maintain with what they bought. To mine multi-predicate

association rules like age(X, "20-29"), occupation(X, "student") purchases(X, "laptop"), we must treat each database attribute or warehouse dimension as a predicate.

Multidimensional association rules are those that involve three or more dimensions or predicates. The rule has three unique predicates: age, occupation, and purchases. For this reason, we may confidently declare that it contains no recursive predicates. Interdimensional association rules span many dimensions but have no common predicates. Multi-dimensional association rules with repeated predicates can also be mined for helpful information. These rules have predicates that appear more than once. Hybrid-dimensional association rules are a term for these types of guidelines. For instance, consider the rule age(X, "20...29") > buys(X, "laptop") > buys(X, "HP printer"), where each iteration of the predicate buys is applied to a different X.

Attributes in a database can be either nominal or quantitative. Nominal attributes (categorical attributes) have "*names of objects*" as their values. There is no hierarchy among the potential values for nominal properties (e.g., occupation, brand, color).Age, salary, and cost are all examples of quantitative qualities, implying a hierarchy of relative importance.

Multidimensional Association Rules in Data Mining Can be Divided Into Two Broad Camps Based on Quantitative Features:

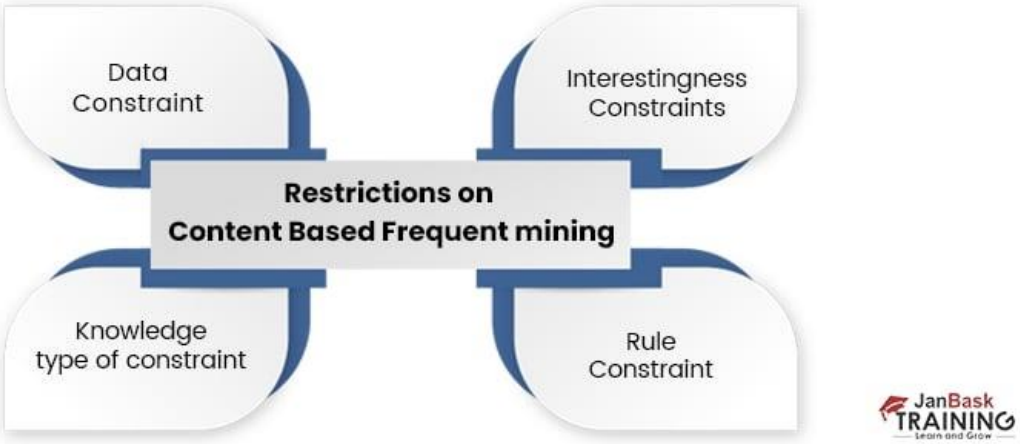
- One method involves leveraging existing idea hierarchies to discretize quantitative features. There is a discretization step before mining. In the case of income, for instance, interval labels such as "0..20K," "21K..30K," "31K..40K," and so on can be used in place of the original numeric values of this attribute thanks to a concept hierarchy. The present case involves a fixed and specified discretization. The interval labels on the discretized numeric attributes allow them to be interpreted as nominal attributes (where each interval represents a category). Using a static discretization of quantitative characteristics, we refer to this as mining multidimensional association rules.
- The second method divides quantitative characteristics into discrete groups, or "*bins*," according to how the corresponding data is distributed. During the mining process, these containers may be mixed together again. For example, a dynamic discretization procedure has been devised to maximize the reliability of the rules mined. Association rules discovered using this method are sometimes known as (vibrant) quantitative association rules. Then numerical attribute values are treated as quantities rather than as established ranges or categories. Let's look into each of these methods for mining MDA rules.

Constraint-Based Frequent Pattern Mining

Data mining allows for extracting thousands of rules that appear to be important from a dataset; nevertheless, likely, the majority of these rules will not provide customers with any value. Users often have a clear concept of the "*form*" of the patterns or rules they want to discover and the

"*direction*" of mining that may lead to intriguing pattern discoveries. It's also possible that they have a preconceived understanding of what the "*conditions*" of the rules are, which would prohibit them from seeking rules that they already know are irrelevant to the situation. As a result of this, a useful heuristic is to have users pick constraints based on their own intuition or preconceptions about what should be allowed. This approach is referred to as "*constraint-based mining*," which is an industry term. The following are some illustrative examples of potential restrictions:

- **Knowledge Type of Constraint:** These characteristics, which may include association, correlation, categorization, or grouping, describe the nature of the knowledge that is to be mined.
- **Data Constraint:** These characteristics are used to determine the information that is required to finish a job. These constraint attempts can be guided in the right direction by imposing constraints, such as limitations on the dimensions or layers of the data, abstractions, or thought hierarchies.
- **Interestingness Constraints:** Limitations on interestingness are utilized in the process of establishing minimum and maximum values for statistica measures of rule interestingness, including support, confidence, and correlation. Limitations are placed on the interestingness of rules.
- **Rule Constraint:** The form or needs of the rules are outlined by the constraints that are placed on the rules to be mined.



In the form of metarules, one can express limitations placed on the number of predicates permitted to occur in the antecedent or consequent of a rule, as well as the relationships between attributes, attribute values, and/or aggregates (rule templates).It is necessary to have both a graphical user interface and a high-level declarative data mining query language to be able to express such constraints.

The first four kinds of limitations have each received a substantial amount of attention over the entirety of this book and in the chapters that came before this one. In this section, we will discuss how the application of rule limits may assist in reducing the overall scope of the mining process. This constraint-based mining approach optimizes the data mining process by allowing users to describe the rules they want to uncover and then look for those rules. In addition, the user-

specified limits can be utilized by an intelligent mining query optimizer, which in turn increases the mining operation's efficiency.

When using constraints, it's possible to do interactive exploratory mining and analysis. In this course, you'll learn about metarule-guided mining, a technique in which syntactic rule restrictions are described using rule templates. details the use of data space pruning (removing portions of the data space for which further exploration cannot contribute to finding patterns matching the requirements) and pattern space pruning (removing portions of the pattern space that are not being mined).

We present anti monotonicity , monotonicity, and succinctness as classes of traits that aid in pruning pattern spaces via constraint-based search space reduction. Convertible constraints are discussed; they are a subset of monotonic and anti-monotonic constraints that may be pushed farther into the iterative mining process without losing their pruning power with the right data ordering.We investigate how data space pruning might be included in a data mining workflow by introducing two sets of properties: data succinctness and data anti monotonicity .

We will assume the user is looking for association rules for the sake of discussion. Adding a correlation measure of interestingness to the support-confidence framework makes it simple to apply the proposed methods to mining correlation rules. For a better understanding of constraint-based frequent pattern mining, you need to learn about the six stages of data science processing.

Metarule-Guided Mining of Association Rules

The user can specify the syntactic form of rules that the user is interested in mining when using a metarule. The mining process's efficiency may be helped by utilizing the rule forms as limitations, which can be done. Metarules can be developed manually by the analyst based on their prior knowledge, expectations, or intuition in relation to the data, or they can be generated automatically depending on the schema of the database.

EXAMPLE : Extracting with a Meteorite as a Guide

Assume you are a market analyst for AllElectronics and have access to both a list of customer transactions and demographic information about the company's clientele (such as age, residence, and credit score). You want to learn if there is a correlation between certain consumer characteristics and the products they buy. You want to know which combinations of client characteristics boost the sale of office software rather than discovering all the association rules expressing these associations. You may use a metarule to define the type of rules you want to uncover.

Metarules like **P1(X, Y) ∧ P2(X, W) ⇒ buys(X, “office software”)** are examples of such expressions, where P1 and P2 are predicate variables that are instantiated to attributes from the given database during mining, X is a variable representing a customer, and Y and W take on values

of the attributes assigned to P1 and P2 during the process. To examine P1 and P2 instantiations, a user will often provide a set of properties. Otherwise, a predetermined set may be applied.

In most cases, a metarule usually generates a working hypothesis about the relationships the user wishes to test or verify. As a result, the data mining software may look for a set of guidelines that correspond to the provided metarule. For example, the Rule is in agreement with the Metarule expression:

age(X, "30..39") ∧ income(X, "41K..60K") ⇒buys(X, "office software").

Let's pretend we're interested in mining association rules across dimensions, like in-

Example 1. Specifically, a Metarule is a Rule Template with The Form P1 ∧ P2 ∧ ⋯ ∧ Pl ⇒ Q1 ∧ Q2 ∧ ⋯ ∧ Qr,

Where Pi (i = 1,..., l) and Qj (j = 1,..., rare instantiated predicates or predicate variables. Let p = l + r represent the total number of predicates in the metarule. Finding all common p-predicate sets, Lp, is necessary for discovering inter-dimensional association rules that fit the template.

In order to calculate the confidence of rules derived from Lp, we additionally need the support or count of the l-predicate subsets of Lp.

In this situation, multidimensional association rule mining is often applied. It is possible to construct effective strategies for metarule-guided mining by extending these approaches with the constraint-pushing techniques discussed below.

1) Reducing The Size of The Pattern Space and The Data Space in Order to Generate Constraint-Based Patterns

Rule constraints can take many different forms. Some examples of rule constraints are the projected set/subset connections of the variables in the mining rules, constant initiation of variables, and constraints on aggregate functions. Users frequently depend on their expertise with the program or data to define rule constraints for the mining work. This is done in order to prevent the mining task from becoming too complex. You can use a more conventional mining technique known as a metarule, or you may use these rule constraints instead of a metarule. In this piece, we take a look at the possible benefits in productivity that might result from incorporating rule limits into your mining operation. To begin, we will look at a case study of hybrid-dimensional association rule mining, which employs rule limitations.

During the mining process, an effective frequent pattern mining processor can narrow its focus by eliminating irrelevant patterns or discarding irrelevant data. The former determines whether or not a pattern may be trimmed by analyzing potential patterns. The Apriori attribute is used to eliminate a pattern if it cannot be further refined by mining into a super pattern . The latter examines the dataset to see if the specific data item may help generate satisfiable patterns (for a certain pattern) in the remaining mining phase. In that case, the information is omitted from further analysis. The

term "*pattern pruning constraint*" refers to a constraint that can be used for pruning in the pattern space. In contrast, "*data pruning constraint*" refers to a constraint that can be used for pruning in the data space.

2) Pattern Space Pruning Through The Use of Pattern Pruning Constraints

There are five kinds of pattern mining restrictions, and each one is based on how a constraint may interact with the process of pattern mining.

- Antimonotonic
- Monotonic
- Concise
- Convertible
- Inconvertible

3) Data Pruning: Reducing Data Volume Constraints for Pruning

In constraint-based frequent pattern mining, pruning data space is the second method of search space reduction. If a piece of data is not going to help generate satisfactory patterns in the mining process, it is removed. Shortness and non-monotonicity of data are two characteristics we take into account.

When employed at the outset of a pattern mining process to prune the data subsets that do not fulfill the constraints, constraints are considered data-succinct. Suppose a mining query specifies that the mined pattern must include the word "*digital camera*," for instance. In that case, any transactions that do not include "*digital camera*" may be removed from the dataset before mining even begins.

Intriguingly, many constraints are data anti-monotonic, meaning that they allow for the elimination of data entries that fail to conform to the present pattern during mining. Due to the fact that it cannot aid in the creation of a super pattern of the present pattern throughout the remaining mining operation, we prune it.

Advantages of Constraint-Based Mining

There are several benefits associated with using constraint-based methods for frequent pattern discovery:



- **Increased Efficiency** - By incorporating user-defined constraints into the algorithm design process, it's possible to reduce search space significantly.
- **Improved Accuracy** - Constraints help eliminate irrelevant or redundant results leading to more accurate findings.
- **Flexibility** - Users can customize their analysis according to their needs by defining different types of constraints.
- **Better Interpretability** - The use of additional information through constraints helps make discovered patterns more interpretable and useful than those obtained using only statistical measures like support count or confidence level.

Pattern Exploration and Application

A pattern repeats itself in response to a set of conditions or regulations. Identifying patterns leads to essential data, simplifies decision-making, and simplifies our tasks. Various parallel missions, such as data management, preprocessing, data warehousing, pipelining, and loading of data for a specific purpose, the creation of machine learning, artificial intelligence models, analysis of the data using these models and existing statistical methods, pattern discovery, and pattern analysis, and finally displaying the facts uncovered, etc., are carried out under the hood of data mining.

Difference Between Data Exploration And Data Mining

Ironically, two primary techniques are mainly used to recover relevant data from gigantic, messy pools. They are known as manual and automatic methods. The main difference between data exploration and data mining is that the manual process is another term for data exploration, while the automated process is known as data mining. Now, Let's study both approaches in detail and compare their differences.

What is Data Exploration?

Data exploration is the initial stage in data analysis. Data analysts utilize data visualization and statistical methods to describe dataset descriptions, such as size, quantity, and accuracy, to understand the data's nature better. Data analysts must first apprehend and create a comprehensive

idea of the data before extracting relevant data for additional analysis. Data exploration strategies include manual analysis and automated data exploration software explanations.

Importance of Data Exploration

It's a well-known fact that humans process visual data more than numerical data. Therefore, it is difficult for data scientists and analysts to give meaning to thousands of rows and columns of data attributes and convey that meaning without any visual elements.Therefore, performing the initial data exploration phase allows data analysts to understand better and visually recognize irregularities and relationships that might otherwise go hidden.

Use Cases of Data Exploration

Data Exploration has been widely operated in multiple industries benefiting different goals. Let us have a look at some use cases of data exploration:

- **Lead Generation:** Companies can extract data from manuals like Yelp, Crunchbase, and Yellowpages and develop leads for business expansion.
- **Content & News Aggregation:** Content aggregation websites can earn regular data feeds from numerous sources and keep their sites new and up-to-date.
- **Sentiment Research:** After extracting the online reviews, comments, and feedback from social media websites like Instagram, Facebook, and Twitter, people can identify the underlying perspectives and understand how they perceive a brand, product, or phenomenon.

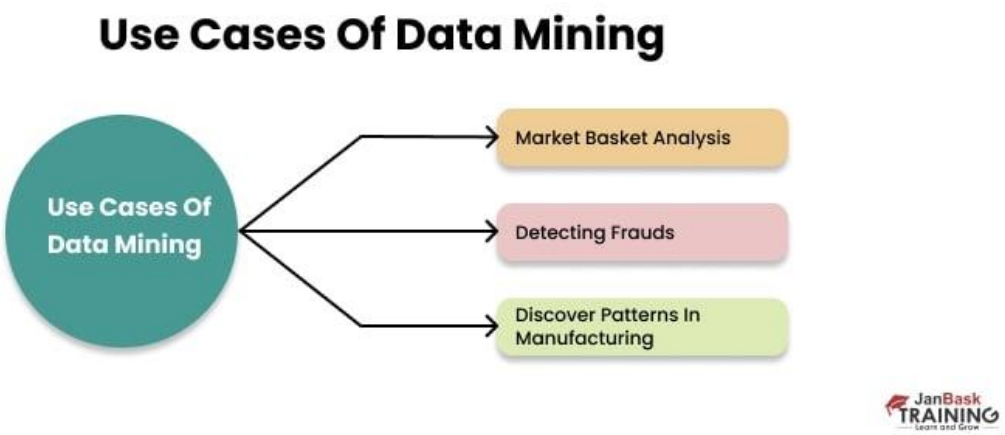
What is Data Mining?

Data mining is an organized and straight method of specifying and finding hidden patterns and data throughout a large dataset. Moreover, it makes machine learning models that are further used in artificial intelligence (AI). Data mining tools can clean through the databases and recognize invisible patterns efficiently by automating the mining procedure. For businesses, data mining is often used to uncover patterns and connections in data to help make optimal business conclusions.

Importance of Data Mining

It is a procedure of capturing large sets of data to define the insights and ideas of the data. It helps to create intelligent market decisions, run objective campaigns, and make predictions. With the help of Data mining, it can study customer behavior and their insights. This results in great success and data-driven business.

Use Cases of Data Mining



Data mining can help segment customers, detect scam activities, forecast sales, etc. Typical data mining uses include:

- **Market Basket Analysis:** This is a technique based on the idea that you are likely to buy another group of products if you buy a particular group. One famous example is that when fathers purchase diapers for their infants, they buy beer together with the diapers.
- **Detecting Frauds:** Data mining assists in structuring models to catch fraud. Organizing samples of fraudulent and non-fraudulent reports empowers businesses to identify suspicious dealings.
- **Discover Patterns in Manufacturing:** In the manufacturing enterprise, data mining is used to support design systems by revealing the relationships between product architecture, portfolio, and customer needs. It can also indicate future product development time and costs.

Types of Data Exploration In Data Mining

We can categorize data exploration types and techniques into manual data exploration and automated data exploration:

Regarding manual data exploration approaches, businesses have a few different options. They can write scripts to analyze raw data using open-source tools built using Python or manual data exploration tools like Microsoft Excel or Google Sheets spreadsheets to examine data in its basic format and create simple charts and data visualizations to detect patterns and correlations between categorical variables. Manual data exploration techniques contain writing scripts to observe raw data or manually filtering data into spreadsheets. Automated data exploration devices, such as data visualization software, benefit scientists by easily monitoring data sources and extensively exploring otherwise massive datasets. Additionally, graphical data displays, such as bar charts and scatter plots, are helpful tools in visual data exploration.

Application of Pattern Exploration In Data Mining

The application pattern discovery in pattern exploration or mining has several uses, which are as follows:

- Pattern mining is commonly used as preprocessing in a variety of data-intensive applications for the removal of noise and data cleaning.
- It can analyze microarray data with tens of thousands of dimensions.
- Pattern mining allows for uncovering hidden algorithms and clusters in data.
- Trees, graphs, subsequences, and networks have all been utilized in pattern mining to examine sequence or structure data.
- Researchers in software engineering use coherent consecutive or gapped subsequences in code execution as sequential patterns to help uncover software faults.

Role of Pattern Exploration In Decision-Making

In this digital age, patterns are everywhere. A pattern can be seen physically or quantitatively through algorithms, apart from the typical mathematical computations such as totals, correlations, variations, ratios, and derivatives. A pattern indicates some important details waiting to be extracted from the data. Among other facts, a pattern could represent a growing trend, a receding factor, the ups and downs of a specific action, seasonal changes, and so on.

Pattern recognition attempts to replicate the neural network abilities of the human brain, which advances artificial intelligence. Pattern recognition is one of the four foundations of computer science. Many practical computer science problems require pattern recognition to be effective. Pattern recognition is the essence of wisdom because patterns embody structure and order, which aid in organizing and making our work more accessible. Identifying and comprehending patterns is a vital part of problem-solving and mathematical reasoning.

Pattern Exploration is Essential in Data Mining:

- It finds and predicts even the most minor bits of hidden or untraceable data; it helps classify unseen data and provides good predictions using learning techniques.
- It can recognize and identify objects from a variety of distances.
- It can assist in developing realistic, actionable predictions based on previously unseen facts.
- It can create a comprehensive knowledge of the data as a crucial fundament for effective and efficient data science efforts.
- Because problems share patterns, we can apply the same problem-solving method everywhere the pattern exists, making them easier to tackle.

How Does Pattern Exploration In Data Mining Or Data Science Work? And Its Types

A pattern repeats itself in response to conditions or regulations. Associating the patterns leads to the required knowledge, makes decision-making more accessible, and simplifies our tasks. Here are some pattern examples that can be found in data mining:

- **Periodic Patterns:** These patterns are repetitive after a given time has passed. This pattern is typical in time series data, biological sequences, spatiotemporal (having both time and spatial dimensions) data, and so on.
- **Associative Patterns:** These are similar to bread and butter" or "knife and fork in that they are a co-occurring group of things that make more sense. The constituents of this pattern are mutually beneficial.
- **Abnormal Patterns:** These patterns appear when the data shows an apparent break from normal behavior, when an unexpected pattern appears between expected pattern, and when its appearance is not periodic.
- **Patterns of Structure:** such as pathfinding in graphs or cluster identification: As an example, low-cost homes are more common in the suburbs, but higher-cost apartments are more common downtown.
- **Chaos Patterns:** These occur but have no defined time, space, or frequency characteristic.

Different Pattern Exploration In Data Mining

Pattern exploration in data mining and analysis is highly beneficial in the following areas:

- **Sentiment and Trend Analysis:** Patterns in this area can provide insight into the sentiments or moods of your customers or investors. If the feeling is positive or negative, this could lead to an increase in demand and price. You can also discover patterns, such as if something is popular or becoming unpopular. This application is widely used in e-commerce, retail, investment banking, voting patterns, and other fields.
- **Image processing and computer vision:** A computer is given human-like abilities to distinguish things by processing their external characteristics, orientations, and behaviors.
- **Bioinformatics:** Massive and complicated biological datasets can be examined to learn about drug side effects, the onset of a pandemic, demographic shortfalls, the genetic foundation of diseases, or changes at the molecular or cellular level.
- **Forensic Analysis and DNA Sequencing:** Recognizing someone's biological parents or identifying a suspect based on remnants of his hair/sputum/skin, etc.
- **Speech Recognition:** This method recognizes information that has been said. The program focuses on a pattern recognition algorithm that identifies spoken text and converts it into ASCII letters displayed on the screen. This also lets us identify the speaker's identity.
- Automation of Assistance and Support: Assisting people and providing better assistance through technology by automating unnecessary daily chores allows employees to focus more on the important ones, thereby increasing employee productivity.
- **Manufacturing:** In this, 3-D images such as structured light, laser, stereo, and so on are used as input, and the objects may be recognized.
- **Character Recognition:** Both letters and numbers are identified by this application. This process takes an optically scanned image as input and produces alphanumeric characters as output. Its primary application is automation and information management. It is also used in page readers, zip codes, registration plates, and other applications.
- **Fingerprint Identification:** In this technique, the input image is obtained through fingerprint sensors, which allows us to identify the fingerprint owner by getting various fingerprint classes.

- **Industrial Automation:** At this point, we deliver the intensity or range image of the product and identify whether it's broken or not.
